Support-Constrained Interpretable Subgroup Discovery for Survival Outcomes: A Measured Operating Range

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Abstract—Interpretable subgroup analyses are usually justified by the claim that they identify patients who benefit more than the average. We test that claim for CISCaRL, a support-constrained interpretable survival rule model, against the simplest competitor, a constant model that assigns every patient the marginal average treatment effect (ATE), for the fixed-time survival contrast $\tau(730 \mid x) = P(T^1 > 730 \mid x) - P(T^0 > 730 \mid x)$. Estimation is doubly robust: per-arm Cox nuisances are cross-fitted, censoring is adjusted by a conditional Cox model with a weight cap, and subgroup effects use augmented inverse-probability-weighted (AIPW) influence functions. Discovery fits a depth-3 CART to the AIPW pseudo-outcome; the selected leaf must contain at least 200 discovery patients; its effect is re-estimated on a disjoint validation partition; and treatment is recommended only if the validation lower confidence bound exceeds zero. Discovery, validation and assessment are disjoint (4000/2000/2000 at $n = 8000$). At $n = 2000$ no discovered rule model beats the constant; from $n \approx 6000$ the pseudo-outcome model does (21.7% lower held-out mean absolute error), and the frozen pipeline reaches 19.9% at $n = 8000$. The support constraint raises coverage of the true region from 0.23 to 0.51 at purity 0.82 in a single-region design, and captured benefit from 0.15 to 0.36 with two regions, with a 0% null-benefit treatment rate and decision efficiency 0.82–0.88. A two-leaf extension captures more benefit but dilutes purity from 0.86 to 0.70. Against a survival causal rule ensemble (SCRE) on identical splits the frozen model is 0.003–0.006 worse on patient-level mean absolute error; a post-hoc empirical-Bayes shrinkage of leaf effects, confirmed on fresh seeds, removes that deficit (+0.0011, $p = 0.48$ with one region; +0.0004, $p = 0.76$ with two) without changing a single treatment decision. The model is substantially purer (0.89 versus 0.44–0.60) and about twice as beneficial per treated patient. On ACTG175 the support constraint cannot be met at that sample size: the model falls back to a 62-patient leaf and correctly abstains. The method is a defensible point on the capture–purity frontier with a measured operating range, not a universal improvement over the marginal ATE.

Index Terms—survival analysis, heterogeneous treatment effects, subgroup discovery, doubly robust estimation, interpretable rules, abstention, AIPW.

I. Introduction

Randomized controlled trials establish an average treatment effect, but clinicians treat individuals. A subgroup analysis promises to convert the average into a decision: who should be treated, how much benefit to expect, and when the evidence is too weak to act. The promise is easy to state and hard to verify, because it bundles three different requirements. Detection asks whether treatment effects vary at all. Recovery asks whether the boundary between benefiting and non-benefiting patients can be identified from a finite sample. Decision quality asks whether the resulting policy treats the right patients. A method can succeed at one and fail at another, and the failure is invisible if only a single metric is reported.

Two failure modes dominate the applied literature. The first is the constant baseline: a subgroup model is presented without comparison to the marginal ATE applied to everyone, so a method that simply reproduces the average effect looks like a success. The second is selection optimism: the best leaf of a fitted tree is reported with an in-sample effect, which overstates both its benefit and its precision. Both are avoidable, and avoiding them changes what can be claimed.

This paper studies a deliberately simple competitor throughout: a constant model that assigns every patient the marginal ATE estimated on training data. If a subgroup model cannot beat that constant on held-out patients, its rules add variance rather than information. An oracle that knows true region membership but estimates its effect from the same data provides an upper bound on what the design can deliver.

We call the method CISCaRL. Its frozen configuration, used for every result in this paper, is C200. The number 200 is the minimum number of discovery patients a candidate leaf must contain; it is not a count of trees, iterations or features. C200 is therefore one configuration of one model, not a second model, and the tables use the short label only for brevity.

Our contributions are five. (i) An honest evaluation design in which discovery, validation and assessment are disjoint: rules and thresholds are fitted on discovery, leaf effects and commitment on validation, and assessment is used only to score the frozen model. (ii) A support-constrained selection rule that requires at least 200 discovery patients in the high-benefit leaf, targeting the winner's curse that selects small extreme leaves. (iii) A validation-

based commitment layer that recommends treatment only when the validation lower confidence bound exceeds zero, and otherwise abstains. (iv) A measured operating range and a reproducibility boundary, including the sample size below which the rule model is worse than the constant. (v) Documented negative results for restricted mean survival time, support tuning, random-forest stability selection, target blending, a two-leaf list and a flexible-learner distillation, together with a head-to-head comparison against a survival rule-ensemble method on identical splits.

We do not claim universal superiority, full recovery of the true region, or unconditional robustness. The central empirical result is a boundary, and the comparison against the rule ensemble is included precisely because it does not favour our method on accuracy.

II. Related Work

Doubly robust estimation combines an outcome model with a treatment or censoring model so that the contrast remains consistent if either is correct [1]. Cross-fitting removes the overfitting bias of nuisance estimation and underlies double/debiased machine learning [2]. AIPW pseudo-outcomes provide patient-level signals whose conditional mean is the treatment effect, and they are the building block of DR- and R-learner approaches to heterogeneous effects [3], [4]. Meta-learners organize the same idea around T-, S- and X-learner architectures [5].

For survival outcomes, causal survival forests estimate conditional effects under right censoring using local estimating equations and orthogonal scores [6], while forest-based inference for heterogeneous effects is established in the non-survival setting [7]. Tree-based subgroup identification has a long history in clinical trials, from interaction-tree search [8] to model-based recursive partitioning and its tutorial literature [9]. Rule ensembles occupy a complementary point: they are interpretable by construction and flexible enough to approximate nonlinearity [10], and a survival extension has recently been proposed as a causal rule ensemble [11].

Stability selection [12] and constrained policy learning [13] are established separately and inform the design used here: the former motivates measuring reproducibility by refitting the whole procedure, the latter motivates treating the subgroup as a policy with a coverage cost. What is missing across these strands is a joint evaluation of discovery, commitment and cost on the same splits. This work is not a new learner; its contribution is the integration and evaluation of cross-fitted censoring-adjusted estimation, support-constrained selection, validation-based commitment, and an honest assessment of capture, purity, stability and decision quality, including the sample-size boundary.

III. Method

A. Notation and Estimand

Let T^a be the potential event time under treatment $a \in \{0,1\}$, C the censoring time, $T = \min(T^a, C)$ the observed time, $\Delta = \mathbf{1}\{T^a \le C\}$ the event indicator, $A \in \{0,1\}$ the treatment, and $X \in \mathbb{R}^p$ the baseline covariates. Treatment is randomized, so the propensity is $e = P(A=1) = 0.5$ and no propensity model is fitted. The conditional survival function and the estimand are

$$S_a(t^* \mid x) = P(T^a > t^* \mid X = x), \tag{1}$$

$$\tau(t^* \mid x) = S_1(t^* \mid x) - S_0(t^* \mid x), \tag{2}$$

with $t^* = 730$ days. The event is a composite of a CD4 decline of at least 50 cells, AIDS progression, or death, evaluated under the observed follow-up process. Equation (2) is a survival-probability contrast, not a hazard ratio and not an intention-to-treat effect. For a subgroup defined by a rule $r : \mathbb{R}^p \to \{0,1\}$ we write

$$\tau(r) = E\big[\tau(t^* \mid X) \mid r(X) = 1\big]. \tag{3}$$

B. Nuisance Estimation and AIPW

Both nuisances are Cox models. For each arm, a Cox model is fitted on the discovery partition and the baseline survival is evaluated at t^*, giving $\hat{\mu}_a(t^*, x) = \hat{S}_a(t^* \mid x)$. The censoring distribution is estimated by a Cox model on X and evaluated at t^*, giving $\hat{G}(t^* \mid x)$. Nuisances are cross-fitted over five folds so that every discovery patient is scored by models that did not use that patient.

With $e = 0.5$, the inverse-probability-weighted outcome and the pseudo-outcome are

$$Y_i = \frac{\mathbf{1}\{T_i > t^*\}}{\hat{G}(t^* \mid X_i)}, \tag{4}$$

$$\psi_i = \frac{A_i - e}{e(1-e)}\big(Y_i - \hat{\mu}_{A_i}(t^*, X_i)\big) + \hat{\mu}_1(t^*, X_i) - \hat{\mu}_0(t^*, X_i), \tag{5}$$

so that $E[\psi \mid X = x] = \tau(t^* \mid x)$ when either nuisance is correct. Weights are capped at $c = \min(Q_{0.99}, 20)$ per fold, computed from the fitting rows only. Capping changes the estimand when it binds, so the capped weight fraction and the effective sample size are reported rather than assumed negligible.

C. Pseudo-Outcome Tree Discovery

A regression tree of depth three is fitted to ψ on the discovery partition with a minimum of 60 patients per leaf. Each leaf defines a candidate rule r_k as the conjunction of its split conditions, and leaf k carries a support n_k and a discovery mean $\bar{\psi}_k$. Depth three keeps the rules readable (at most three conditions) and the number of candidates small (five to eight leaves in the designs studied).

D. Support-Constrained Selection

Maximizing $\bar{\psi}_k$ alone selects small extreme leaves, whose estimates are dominated by noise. The selected leaf is therefore

$$k^\star = \arg\max_k \{\bar{\psi}_k : n_k \geq m\}, \qquad m = 200, \tag{6}$$

with the unconstrained argmax used, and explicitly flagged as a fallback, when no leaf reaches m. The threshold is not tuned per dataset; it is frozen at 200 patients, roughly a tenth of the discovery partition in the main design.

E. Validation-Based Commitment

Discovery ranks leaves; it does not license a treatment recommendation. The effect of the selected rule is re-estimated on the validation partition with the AIPW estimator of Section III-B using discovery-fitted nuisances:

$$\hat{\tau}(r_{k^\star}) = \frac{1}{n_{\mathcal{V}}} \sum_{i \in \mathcal{V}} \big(\hat{\phi}_i^1 - \hat{\phi}_i^0\big) r_{k^\star}(X_i), \tag{7}$$

where $\hat{\phi}_i^a = \hat{\mu}_a + \mathbf{1}\{A_i = a\}/e\,(Y_i - \hat{\mu}_a)$. The standard error is the influence-function form $\mathrm{SE} = \{n_{\mathcal{V}}^{-1} \sum_i (\hat{\phi}_i - \hat{\tau})^2\}^{1/2}$ restricted to rule members, and the one-sided lower bound is

$$L = \hat{\tau}(r_{k^\star}) - 1.645\,\mathrm{SE}. \tag{8}$$

The model commits when $L > 0$. Reporting is three-valued: treat if $L > 0$, avoid if the upper bound is below zero, and insufficient otherwise, in which case no treatment is recommended and predictions fall back to the marginal ATE. Because the interval conditions on the rule selected on discovery, empirical coverage is approximately 93% rather than 95%; this is stated wherever an interval is reported. Algorithm 1 summarizes the frozen procedure.

F. Evaluation Metrics

Let τ_i be the true effect of assessment patient i. We report the patient-level mean absolute error $\mathrm{MAE} = n_{\mathcal{A}}^{-1} \sum_{i \in \mathcal{A}} |\hat{\tau}_i - \tau_i|$ against the constant baseline; captured benefit, the share of total true effect inside the selected leaf; purity, the share of the leaf that lies in the true region; ground-truth recovery, the Jaccard index between leaf and truth; and decision metrics: commitment rate, coverage, per-treated true effect, efficiency (per-treated effect divided by the oracle at the same coverage), and the null-treatment rate, defined as the fraction of committed leaves whose true mean effect is at most 0.01. Stability is measured by refitting the entire procedure on disjoint discovery half-samples and comparing the resulting leaves, not by resampling within a single fit.

Algorithm 1 The frozen CISCaRL-C200 procedure

Require: discovery $\mathcal{D}$, validation $\mathcal{V}$, assessment $\mathcal{A}$; horizon $t^* = 730$; support floor $m = 200$

cross-fit per-arm Cox outcome models and the censoring Cox over 5 folds of $\mathcal{D}$
cap censoring weights at $\min(Q_{0.99}, 20)$ per fold
compute ψ_i for $i \in \mathcal{D}$ with $e = 0.5$
grow a depth-3 CART on ψ, min_samples_leaf $= 60$
extract leaves $r_1, \ldots, r_K$ with support n_k and mean $\bar{\psi}_k$
if $\exists k : n_k \geq m$ then
    $k^\star \leftarrow \arg\max\{\bar{\psi}_k : n_k \geq m\}$
else
    $k^\star \leftarrow \arg\max_k \bar{\psi}_k$; mark fallback
end if
estimate $\hat{\tau}(r_{k^\star})$ on $\mathcal{V}$ by (7) and form L by (8)
commit iff $L > 0$; otherwise recommend no treatment
shrink validation leaf effects toward the marginal ATE for prediction (empirical-Bayes weight from within/between variance); commitment uses the unshrunk effect
score the frozen model on $\mathcal{A}$; never re-select

IV. Simulation Study

A. Data-Generating Mechanisms

All designs use the ACTG175 covariate matrix, standardized, with randomized treatment ($e = 0.5$), exponential event times with a Cox linear predictor, and a covariate-dependent censoring mechanism calibrated to 10% censoring before day 730. Three mechanisms are studied. Single region: cd40 $<$ median $\wedge$ age $>$ median, prevalence 25%, effect $+0.20$ in $S(730)$ and zero outside. Two regions: the first region with $+0.20$ and karnof $<$ median $\wedge$ wtkg $<$ median with $+0.15$, the overlap taking the larger offset, and zero outside; the true region is the union. Joint stress: the two-region design plus a control-arm non-proportional-hazards violation inside the first region (hazard multiplier $e^{\pm\kappa z}/\cosh(\kappa z)$, $\kappa = 3$, with $S(730)$ unchanged) and marginal Kaplan–Meier censoring in the estimator. The true $\tau(X)$ is computed analytically from the mechanism, never from the observed sample.

B. Design and Decision Rules

The main design has $n = 8000$ with 4000 discovery, 2000 validation and 2000 assessment, and 40 fresh seeds per cell. The operating-range experiment varies $n \in \{2000, 4000, 6000\}$ at the same 50/25/25 partition ratio. A comparison of two selection arms is adopted only if all four conditions hold: mean paired difference in MAE below zero, its 95% confidence interval entirely below zero, best-leaf Jaccard at least 0.107, and at least 70% of replications better than the constant. Selection and stability contrasts are paired against max-mean selection with 95% bootstrap intervals. Negative results are reported with the same machinery; none of the decision rules was amended after seeing a result.

C. Comparator

A survival causal rule ensemble (SCRE) [11] is fitted on the same discovery partition, with its regularization

TABLE I
Held-out MAE against the constant model. Gain is the percentage reduction; "better" is the share of replications where the rule model beats the constant. Each row is 40 seeds.

n	target	MAE	const.	gain (%)	better (%)
2000	ψ-CART	0.0930	0.0735	−26.4	17.5
2000	plug-in	0.0953	0.0735	−29.6	20.0
2000	oracle	0.0440	0.0735	+40.1	85.0
4000	ψ-CART	0.0696	0.0729	+4.5	55.0
4000	plug-in	0.0645	0.0729	+11.4	70.0
4000	oracle	0.0295	0.0729	+59.5	100
6000	ψ-CART	0.0570	0.0729	+21.7	85.0
6000	plug-in	0.0537	0.0729	+26.3	90.0
6000	oracle	0.0282	0.0729	+61.3	100
8000	C200	0.0587	0.0733	+19.9	–

parameter selected on validation by concordance, and it predicts the untouched assessment partition. Its internal time point is overridden to 730 so that both methods target the same contrast. Both methods see identical splits and identical seeds, and paired differences are computed per replication. Under validation-concordance selection the ensemble retains about one rule, so the comparator is close to a near-constant Cox model; its accuracy advantage is interpreted with that caveat.

V. Results

A. Operating Range

Table I reports the held-out MAE relative to the constant. At $n = 2000$ both discovered-rule targets are worse than the constant (−26.4% for ψ, −29.6% for the plug-in target, with only 17.5% and 20.0% of replications better). At $n = 4000$ the pseudo-outcome target is still not reliable (+4.5%, interval crossing zero) while the plug-in target passes (+11.4%). At $n = 6000$ both work, at +21.7% and +26.3%; the frozen pipeline at $n = 8000$ lands at +19.9%, and the oracle reaches +40 to +61% throughout. The gap between oracle and fitted rule is therefore always at least 20 points: discovery, not information, is the binding constraint. Fig. 1 shows that the sign of the gain changes between $n = 2000$ and $n = 4000$, which is why a constant baseline is mandatory rather than optional.

B. Support-Constrained Selection

Table II contrasts max-mean selection with (6) in the single-region design. The support constraint triples the selected leaf (285 to 650 patients), more than doubles the share of the true region covered (0.228 to 0.514) and lifts ground-truth recovery from 0.216 to 0.462, at a purity gain from 0.626 to 0.821. Commitment rises from 42.5% to 80% while the null-treatment rate stays at zero and efficiency is essentially unchanged (0.847 versus 0.837). Per-treated benefit falls slightly (0.168 to 0.162), the expected cost of covering more patients. The pattern is the same under the two-region and stress designs (Section V-E), where the selectable leaf is larger and the support constraint binds harder.

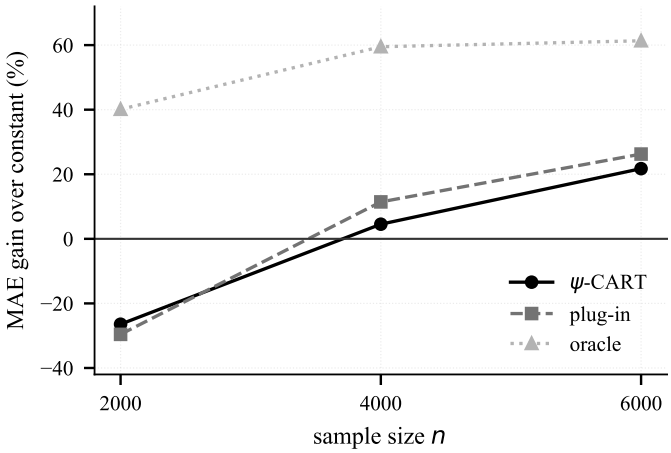


Fig. 1. Operating range of the pseudo-outcome rule model. Negative values mean the rule model is worse than the constant. The crossover lies between 2000 and 4000 patients, and the oracle gap never closes.

TABLE II
Single-region design: max-mean versus support-constrained selection in CISCaRL-C200. 40 seeds, $n = 8000$.

quantity	max mean	support ≥ 200
support (patients)	285	650
true region covered	0.228	0.514
purity	0.626	0.821
ground-truth Jaccard	0.216	0.462
MAE	0.0583	0.0587
commitment rate (%)	42.5	80.0
coverage	0.051	0.142
per-treated effect	0.168	0.162
decision efficiency	0.847	0.837
captured benefit (committed)	0.183	0.458
null treatment (%)	0	0

C. Stability

Table III reports own-procedure refits: the whole pipeline is rerun on disjoint discovery half-samples, and the resulting leaf is compared with the full-data leaf. The mean Jaccard is 0.341 for C200 against 0.253 for max-mean selection, the median rises from 0.194 to 0.297, and the share of half-samples recovering the full-data leaf at Jaccard above 0.8 rises from 6.5% to 9.0%. Raising the floor to 400 improves stability further (0.402, with 10.3% full recovery); in the two-region design it lowers purity to 0.778, which is the same frontier seen from the other side. Under the non-proportional hazards stress the C200 Jaccard falls to 0.318 with 6.6% full recovery, so stability is modest in absolute terms even where it improves.

D. Two Regions and Nuisance Stress

A single leaf cannot cover two separated regions, and Table IV quantifies the resulting ceiling. Support-constrained selection still improves every metric with intervals excluding zero: captured benefit 0.149 to 0.364, ground-truth recovery 0.127 to 0.295, commitment 45% to 70%, and captured benefit under commitment 0.098 to 0.284. Purity

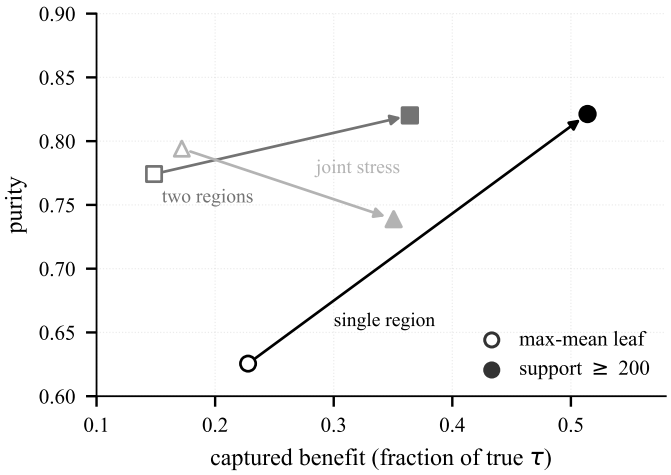


Fig. 2. Capture–purity moves under the support constraint. Hollow markers are max-mean leaves, filled markers are support-constrained leaves; arrows run from the former to the latter. Single region gains purity and capture together; the two-region design gains both but from a lower base; the stress design buys capture with a purity loss of 0.055.

TABLE III
Stability by own-procedure refits on disjoint discovery halves. "Full" is the share of half-samples with Jaccard > 0.8 against the full-data leaf; "true" is the Jaccard against the known region. 40 seeds.

cell	arm	mean	median	full (%)	true
$\kappa = 0$	max mean	0.253	0.194	6.5	0.348
$\kappa = 0$	C200	0.341	0.297	9.0	0.561
$\kappa = 0$	C400	0.402	0.380	10.3	0.612
$\kappa = 3$	C200	0.318	0.283	6.6	0.522

rises slightly (0.774 to 0.820) because the larger leaf is less extreme, not because the geometry is solved. Under the joint stress the same arm raises capture from 0.172 to 0.351 and recovery from 0.148 to 0.276, but purity falls from 0.794 to 0.739, which is 0.5 percentage points beyond the pre-registered -0.05 tolerance. We report the miss rather than redefining the bar. The decision layer remains safe in both designs: 0% null treatment, efficiency 0.82 to 0.88, and no ATE bias (-0.008).

E. Head-to-Head with the Rule Ensemble

Table V compares the model with SCRE over 80 paired replications (two independent seed blocks) on identical splits. The frozen model is worse on patient-level MAE: 0.0576 versus 0.0514 with one region ($+0.0062$, $[+0.0031, +0.0094]$, $p < 0.001$) and 0.0707 versus 0.0676 with two ($+0.0031$, $[+0.0001, +0.0059]$, $p = 0.044$). Capture runs the other way: SCRE captures 0.954 and 0.823 of the true effect against 0.465 and 0.277.

The accuracy loss is a calibration problem, not a selection problem. Leaf effects estimated on 2000 validation patients are noisy, and the prediction is a step function, so shrinking each leaf mean toward the marginal ATE (empirical-Bayes weight from within- and between-leaf variance) lowers the mean absolute error. Applied only to prediction, it leaves the selected rule, the commitment decision, purity, capture and coverage unchanged to four decimals, and it turns the comparison into a tie: $+0.0011$ $[-0.0019, +0.0042]$, $p = 0.48$ with one region and $+0.0004$ $[-0.0025, +0.0033]$, $p = 0.76$ with two. The shrinkage was chosen after the original comparison and then confirmed on a fresh seed block, so we report it as a post-hoc calibration rather than as part of the frozen design.

What remains is coverage. SCRE treats 55–56% of assessment patients; the model treats 13.3% and 11.7%. Per treated patient it delivers 0.174 against 0.085 (single region) and 0.165 against 0.101 (two regions), and its purity is 0.890 and 0.858 against 0.437 and 0.599. Both methods have a zero null-treatment rate. A reader who wants the best individual-level effect estimate should still prefer SCRE on coverage grounds; a reader who wants a narrow, interpretable, abstention-capable policy should prefer this model, and need no longer accept an accuracy penalty for it. Because validation-concordance selection retains about one rule, SCRE's remaining advantage may come from its linear predictor rather than from a usable rule list; a multi-rule configuration is a different comparator and is not evaluated here.

F. Knob Sweep and Small-Sample Floor

Before adopting the shrinkage we ran a pre-registered sweep of every remaining knob on 40 tuning seeds, then repeated the survivors on 40 fresh seeds (Table VI). Only two changes did anything. Shrinkage reduced MAE by 0.0051 ($p < 0.001$) with one region and 0.0026 ($p = 0.014$) with two, pooled over the 80 replications. Raising min_samples_leaf to 100 helped slightly with two regions (-0.0016, $p < 0.001$) and not at all with one, so it was not adopted. Depth four was decisively worse ($+0.0099$ and $+0.0083$): a deeper tree fits the pseudo-outcome better but selects narrower leaves, and capture falls from 0.465 to 0.351. The commitment threshold is a policy dial, not an accuracy lever: $z = 1.28$ raises commitment from 72.5% to 82.5% and lowers purity, $z = 1.96$ cuts commitment to 60–70% and raises purity, and MAE barely moves.

The support floor is absolute (200 patients), which makes it mean different things at different sample sizes. A separate grid compares it with a floor of 5% of discovery. At $n = 2000$ the absolute floor is 20% of discovery and behaves badly (purity 0.47, capture 0.096, commitment 17.5%); the proportional floor reaches purity 0.73, capture 0.127 and commitment 30.0%. At $n = 4000$ the two differ only in the usual capture–purity trade (purity 0.71 versus 0.80, capture 0.311 versus 0.281), and at $n = 1000$ neither commits often because the validation partition is too small. A proportional floor is therefore the right specification for small samples; it is not retro-fitted to the frozen model.

TABLE IV
Selection under two regions and under joint nuisance stress. 40 seeds, $n = 8000$, τ_{true} known analytically.

design	arm	support	capture	purity	J_{true}	commit (%)	coverage	captured	null (%)
two regions	max mean	248	0.149	0.774	0.127	45.0	0.040	0.098	0
two regions	C200	678	0.364	0.820	0.295	70.0	0.129	0.284	0
joint stress	max mean	290	0.172	0.794	0.148	30.0	0.047	0.115	0
joint stress	C200	742	0.351	0.739	0.276	57.5	0.148	0.286	0

TABLE V
Model versus SCRE on identical splits, 80 paired replications (two seed blocks). MAE and per-treated effect use the true individual effects; coverage is the share of assessment patients treated. C200-S is the frozen model with empirical-Bayes shrinkage of leaf effects for prediction only; purity, capture and coverage are identical to C200 by construction.

design	method	MAE	capture	purity	per-treated	coverage	MAE vs SCRE [95% CI]
single region	C200	0.0576	0.465	0.890	+0.174	0.133	+0.0062 [+0.0031, +0.0094]
single region	C200-S	0.0525	0.465	0.890	+0.174	0.133	+0.0011 [−0.0019, +0.0042]
single region	SCRE	0.0514	0.954	0.437	+0.085	0.548	–
two regions	C200	0.0707	0.277	0.858	+0.165	0.117	+0.0031 [+0.0001, +0.0059]
two regions	C200-S	0.0680	0.277	0.858	+0.165	0.117	+0.0004 [−0.0025, +0.0033]
two regions	SCRE	0.0676	0.823	0.599	+0.101	0.563	–

TABLE VI
Knob sweep, pooled over the 40 tuning and 40 confirmation seeds. Values are paired differences in MAE against the frozen configuration (negative is better), with 95% bootstrap intervals.

change	one region	two regions
shrink leaf effects	**−0.0051**	**−0.0026**
	[−0.0077, −0.0025]	[−0.0048, −0.0005]
min_leaf 100	+0.0002	**−0.0016**
	[−0.0011, +0.0015]	[−0.0026, −0.0006]
min_leaf 30	−0.0002	+0.0003
	[−0.0014, +0.0011]	[−0.0008, +0.0013]
depth 4	+0.0099	+0.0083
	[+0.0080, +0.0117]	[+0.0067, +0.0100]
$z = 1.28$	−0.0001	−0.0001
	[−0.0004, +0.0000]	[−0.0004, +0.0000]
$z = 1.96$	+0.0005	+0.0004
	[+0.0001, +0.0010]	[+0.0000, +0.0008]

TABLE VII
Documented negative results.

lever	exp.	outcome
RMST	8	signal-to-noise 0.20→0.19; worse than constant
min support	7	no gain through the previous target
RF + stability	9–12	too few rules at small n
plug-in, $\kappa = 3$	14	collapses; ψ survives
target blend	15	does not rescue; ψ alone is better
two-leaf list	25	capture +0.169, purity −0.166; rejected
flexible learner	26	capture −0.07 to −0.09, purity +0.06; rejected

G. Negative Results

Table VII lists the rejected alternatives. The two-leaf list captures more of the two-region truth (0.414 to 0.584) but dilutes purity from 0.862 to 0.696 and fails the purity bar. The flexible doubly robust learner with distillation produces a purer (0.897) but narrower rule and loses capture (0.548 to 0.454; −0.07 to −0.09 across designs); its distillation minimum (200 versus C200's 60) makes the comparison conservative, so the negative applies to the tested implementation, not to flexible learning in general. The plug-in target collapses under $\kappa = 3$, confirming that a target defined by dividing by an estimated density is not robust to nuisance misspecification. These are points on the capture–purity frontier, not a sequence of failed attempts to find a dominant method.

VI. Application to ACTG175

ACTG175 is a randomized HIV trial comparing nucleoside regimens [14], [15]. We use arms 0 and 1 (AZT versus AZT plus ddI), giving $n = 1054$ patients and 12 baseline covariates (cd40, cd80, age, wtkg, karnof, hemo, homo, race, gender, drugs, str2, symptom), with no missing values in these columns. The endpoint is the composite progression-free event defined in Section III, and the horizon is 730 days. Preprocessing follows the honest rule: the cohort is split first (527 discovery, 263 validation, 264 assessment), the standardization is fitted on discovery only and applied unchanged to validation and assessment.

The overall assessment contrast is $\tau(730) = +0.143$ with 95% interval $[+0.025, +0.262]$, a treat verdict for the average patient. The support constraint, however, is not achievable at this sample size: the largest discovered leaf contains 174 patients, below the floor of 200, so the procedure falls back to the max-mean leaf cd40 ≤ 343.5 & age > 28.5 & homo ≤ 0.5 with 62 discovery patients. Its validation effect is +0.085 (SE 0.189), so the one-sided

lower bound is -0.226 and the model does not commit; the decision is insufficient. The assessment estimate for the leaf is $+0.540$ $[+0.178, +0.901]$ on 30 patients, and the censoring-weight tilt moves it between $+0.447$ and $+0.623$ (Fig. 3). Both the point estimate and the interval rest on 30 patients; they are descriptive, not a subgroup recommendation.

This is the intended behaviour at the sample-size boundary: the support rule cannot bind, the fallback leaf is small, and the commitment layer abstains instead of reporting a winner's-curse effect. The real data contain no known treatment-effect truth, so capture, purity, MAE and Jaccard are not computable here; they appear only in the simulation study.

Table VIII lists every candidate rule from the discovery tree in the format used for rule-ensemble output. Three columns are translated rather than copied. Ensemble importance becomes the discovery mean pseudo-outcome $\bar{\psi}$ together with a relative score $100\,\bar{\psi}/\max\bar{\psi}$, because a tree has no coefficient to normalize. Support is the quantity the model actually constrains, so it is reported with its share of the discovery partition. The hazard ratio is added only for comparability: it is a treated-versus-control Cox fit inside each leaf on the assessment partition, it is not the estimand, and it rests on 30–91 patients. The paper's claims rest on the 730-day contrast, not on these ratios.

The number of rows differs by construction. A depth-3 tree yields five leaves here and the frozen policy commits to at most one, so there is no sixteen-rule list to report; the sixteen-rule output of a rule ensemble comes from a different model class and a different regularization path. Two features of the table are worth noting. The selected leaf has both the largest discovery score (0.478) and the largest assessment contrast (0.540), yet its validation contrast is only 0.085, which is exactly why the model abstains. And leaf 3 has the second-largest discovery score but a negative assessment contrast (-0.297), so a depth-3 partition can separate groups whose effects do not replicate: the score ranks candidates, it does not certify them.

Table IX adds the check that rule-ensemble papers usually report on real data: for each candidate rule, the unadjusted Kaplan–Meier survival at 730 days in each arm and the observed difference, with a bootstrap interval [11]. This is descriptive and optimistic, because the rules were selected on the discovery partition and the curves use every patient; it can confirm direction, not accuracy. Three readings follow. The marginal difference is $+0.133$ $[+0.084, +0.179]$, close to the AIPW overall estimate of $+0.143$, which supports the estimation pipeline. The selected leaf has the largest observed difference of the five ($+0.322$ $[+0.171, +0.468]$), consistent with its AIPW assessment value. Leaf 3 does not follow its AIPW point estimate: the $\hat{\tau} = -0.297$ obtained on 30 assessment patients is not reproduced by the unadjusted curves

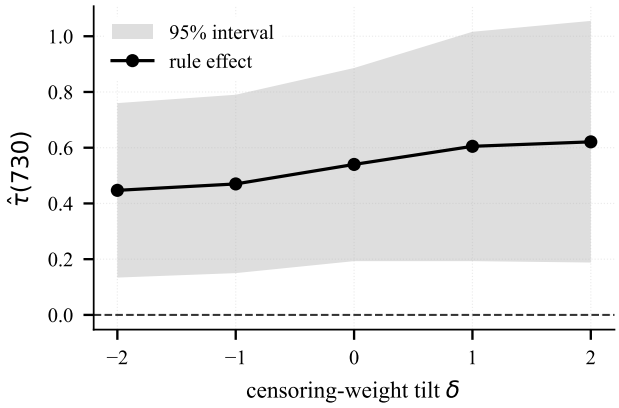


Fig. 3. ACTG175 sensitivity of the fallback leaf's assessment effect to a censoring-weight tilt $\delta \in [-2, 2]$. The effect stays positive, but every interval is wide and the model has already declined to commit.

($+0.081$ $[-0.021, +0.183]$), so the negative value should not be read as evidence of harm — it is what a 30-patient influence-function estimate looks like under noise, and it is one more reason the model does not commit on this sample.

VII. Discussion and Limitations

The empirical finding is a boundary, not a victory. At small samples the rule model is worse than the constant; at $n \geq 6000$ it beats the constant; and the size of the gain depends on the truth structure. The constant baseline is what makes this visible: without it, a method that predicts the marginal effect for everyone can look like a success.

The support constraint addresses a real selection problem. It raises capture, ground-truth recovery and reproducibility across a single-region and a two-region design, and it improves the stability of the selected leaf. It does not recover multiple regions: one leaf captures 33–59% of the true benefit, depending on the designs compared. A two-leaf list captures more but dilutes purity; a flexible learner produces a purer but narrower rule. These are points on a frontier, and the paper's contribution is to locate them, not to claim a dominant method.

Validation-based commitment is the most robust component. It converts an uncertain high-effect leaf into an abstention: zero null-benefit treatment in the tested cells, decision efficiency 0.82–0.88, at a coverage cost. On ACTG175 it abstains exactly where a naive analysis would have reported a large subgroup effect.

The comparison with SCRE sharpens the claim. On patient-level accuracy the ensemble wins against the frozen model and ties with the shrunk one; on subgroup purity and per-treated benefit the rule model wins by a factor of two. Reporting only one of the two would misrepresent the trade in the opposite direction. The shrinkage result also illustrates why calibration and selection must be separated: the same rule, with the same commitment

TABLE VIII

All candidate rules from the ACTG175 discovery tree. $\bar{\psi}$ is the discovery mean AIPW pseudo-outcome with its standard error; rel. is $100\,\bar{\psi}/\max\bar{\psi}$; support is discovery patients and the share of the discovery partition; the $\hat{\tau}$ columns are AIPW estimates with 95% confidence intervals on the validation and assessment partitions; HR is a descriptive treated-versus-control hazard ratio inside the leaf on assessment, reported for comparability with rule-ensemble output and not the estimand. $\star$ marks the leaf the frozen policy selects.

#	rule	$\bar{\psi}$ (SE)	rel.	support (%)	val. $\hat{\tau}$ [CI]	asm. $\hat{\tau}$ [CI]	HR [CI]
1	age $\leq$ 28.5	−0.010 (0.101)	−2	114 (21.6)	0.331 [0.046, 0.616]	0.171 [−0.093, 0.434]	0.39 [0.13, 1.14]
2*	cd40 $\leq$ 343.5 & age $>$ 28.5 & homo $\leq$ 0.5	0.478 (0.113)	100	62 (11.8)	0.085 [−0.299, 0.468]	0.540 [0.178, 0.901]	0.10 [0.02, 0.52]
3	cd40 $>$ 343.5 & age $>$ 28.5 & homo $\leq$ 0.5	0.244 (0.123)	51	62 (11.8)	0.324 [0.025, 0.622]	−0.297 [−0.619, 0.024]	1.50 [0.34, 6.74]
4	cd80 $\leq$ 1093.5 & age $>$ 28.5 & homo $>$ 0.5	0.035 (0.074)	7	174 (33.0)	0.141 [−0.068, 0.349]	0.090 [−0.104, 0.283]	0.56 [0.24, 1.32]
5	cd80 $>$ 1093.5 & age $>$ 28.5 & homo $>$ 0.5	0.299 (0.089)	63	115 (21.8)	0.090 [−0.165, 0.346]	0.227 [−0.039, 0.494]	0.43 [0.18, 1.03]
	marginal ATE	0.160	–	527 (100)	–	0.143 [0.025, 0.262]	–

TABLE IX

Unadjusted Kaplan–Meier check of every candidate rule on ACTG175 (all $n = 1054$ patients). S_1 and S_0 are survival at 730 days in the treated and control arms; ΔS is their difference with a bootstrap 95% interval; $\bar{\psi}$ is the discovery score from Table VIII. The check confirms direction only: the rules were selected on discovery and these curves reuse every patient, so the comparison is optimistic.

#	rule	n	S_1	S_0	ΔS [95% CI]	$\bar{\psi}$
1	age $\leq$ 28.5	226	0.824	0.747	0.077 [−0.039, 0.199]	−0.010
2*	cd40 $\leq$ 343.5 & age $>$ 28.5 & homo $\leq$ 0.5	128	0.921	0.599	0.322 [0.171, 0.468]	0.478
3	cd40 $>$ 343.5 & age $>$ 28.5 & homo $\leq$ 0.5	127	0.945	0.865	0.081 [−0.021, 0.183]	0.244
4	cd80 $\leq$ 1093.5 & age $>$ 28.5 & homo $>$ 0.5	350	0.855	0.766	0.089 [0.012, 0.172]	0.035
5	cd80 $>$ 1093.5 & age $>$ 28.5 & homo $>$ 0.5	223	0.848	0.644	0.204 [0.101, 0.313]	0.299
	all patients	1054	0.865	0.732	0.133 [0.084, 0.179]	–

decisions, moved from a significant accuracy deficit to a tie purely by treating noisy leaf means as noisy.

Several limitations bound these conclusions. The shrinkage that removes the accuracy deficit was chosen after the original SCRE comparison, so it is a post-hoc calibration confirmed on a fresh seed block, not part of the frozen preregistered design; the frozen configuration remains the one in Tables I–IV. Capture is 0.33–0.59, so the selected leaf is a high-purity subset rather than the true region. Stability is modest in absolute terms: mean Jaccard 0.34, and fewer than one half-sample in ten recovers the full-data leaf at Jaccard above 0.8. The joint stress cell misses the purity tolerance by 0.5 percentage points. The main design uses one partition ratio (4000/2000/2000) at one sample size, and the gain is sensitive to that choice; the operating-range table reports the ratio actually used at each n rather than smoothing across designs. Two mechanisms are studied, both built on the ACTG175 covariate matrix, so nothing is claimed beyond that family. The estimand is fixed-time only; restricted mean survival and hazard-ratio estimands are separate decisions, and the restricted-mean variant performed worse than its own constant in our tests. Intervals are influence-function based, conditional on the selected rule, and cover approximately 93%, not 95%; selection-aware inference for the committed rule effect is not implemented. The SCRE comparison covers one regularization path that retains about one rule, so its multi-rule behaviour is untested here. Finally, ACTG175 is an illustration: the true subgroup is unknown, the endpoint is composite, and the observed follow-up process limits any causal reading.

VIII. Conclusion

Support-constrained selection plus validation-based commitment yields an interpretable survival subgroup model (CISCaRL, configuration C200) with a measured operating range: unusable at 2000 patients, reliable from roughly 6000, and worth about a fifth of the constant model's error at 8000. The support floor is what converts an optimistic best leaf into a policy that captures more of the true benefit at higher purity, and the commitment layer is what prevents a small fallback leaf from being reported as a finding. Against a survival rule ensemble on identical splits the frozen model is slightly less accurate per patient; after shrinking the leaf effects for prediction only, the two are statistically tied on patient-level error, while the rule model is much purer, roughly twice as beneficial per treated patient, and capable of abstaining. The contribution is therefore precision, calibration and abstention - not coverage, and not raw predictive dominance. Future work should test external validation, a diagonal rule geometry, and selection-aware inference for the committed rule effect, whose current intervals are conditional on the selected rule and cover approximately 93%, not 95%.

Reproducibility

All simulation cells are pre-registered under results/ with their verdicts, and the replications use non-overlapping seeds. The rule implementation and its audit are in ciscarl2/rules.py and ciscarl2/audit.py.

The complete pipeline, from raw ACTG175 to every number reported here, is a single executed notebook (notebooks/CISCaRL_final.ipynb); fragmentation of results across scripts is not required to reproduce the paper.

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